

We take your input:

```
arr = [6, 3, 9, 5, 2, 8]
```

STEP 1: First call

```
merge_Sort(arr, 0, 5)
```

- si=0, ei=5
- mid = 2

Call stack banega:

```
merge_Sort(0,2)
```

```
merge_Sort(3,5)
```

LEFT HALF (0 → 2)

Call:

```
merge_Sort(0,2)
```

- mid = 1

```
merge_Sort(0,1)
```

```
merge_Sort(2,2)
```

◇ **Call:**

```
merge_Sort(0,1)
```

- mid = 0

```
merge_Sort(0,0) → return
```

```
merge_Sort(1,1) → return
```

NOW MERGE (0,1)

```
merge(arr, 0, 1, 0)
```

Initial:

```
i=0 (6)  
j=1 (3)  
temp = []
```

While loop:

- compare 6 vs 3 → 3 chhota

```
temp = [3]  
j=2  
  
Loop ends (j > ei)
```

Left part:

```
temp = [3,6]
```

COPY BACK:

```
arr[0]=3  
arr[1]=6
```

👉 Array becomes:

```
[3, 6, 9, 5, 2, 8]
```

◇ Call:

```
merge_Sort(2,2)
```

→ return (single element)

MERGE (0,2)

```
merge(arr, 0, 2, 1)
```

Initial:

Left = [3,6]

Right = [9]

Compare:

- $3 < 9 \rightarrow \text{temp} = [3]$
- $6 < 9 \rightarrow \text{temp} = [3,6]$

Right part:

temp = [3,6,9]

COPY BACK:

```
arr = [3,6,9,5,2,8]
```

RIGHT HALF (3 → 5)

Call:

```
merge_Sort(3,5)
```

```
mid = 4
```

```
merge_Sort(3,4)
```

```
merge_Sort(5,5)
```

◇ Call:

`merge_Sort(3,4)`

`mid = 3`

`merge_Sort(3,3) → return`

`merge_Sort(4,4) → return`

✂ MERGE (3,4)

`merge(arr, 3, 4, 3)`

Initial:

5 vs 2

→ `temp = [2]`

→ then 5

`temp = [2,5]`

💧 COPY BACK:

`arr = [3,6,9,2,5,8]`

◇ Call:

`merge_Sort(5,5)`

→ return

MERGE (3,5)

```
merge(arr, 3, 5, 4)
```

Initial:

Left = [2,5]

Right = [8]

Build temp:

2 → 5 → 8

COPY BACK:

arr = [3,6,9,2,5,8] → [3,6,9,2,5,8] (already same)

FINAL MERGE (0,5)

```
merge(arr, 0, 5, 2)
```

Initial:

Left = [3,6,9]

Right = [2,5,8]

Step-by-step:

2 → temp [2]

3 → temp [2,3]

5 → temp [2,3,5]

6 → temp [2,3,5,6]

8 → temp [2,3,5,6,8]

9 → temp [2,3,5,6,8,9]

FINAL COPY BACK:

`arr = [2,3,5,6,8,9]`

THE MOMENT (IMPORTANT)

Tu pooch raha tha:

“copy back kyun?”

 Ye dek:

Har merge ke baad agar copy back nahi hota:

- next merge **old array** use karta
- sorted pieces kabhi combine hi nahi hote

FINAL MENTAL MODEL

Tera code actually ye kar raha hai:

Divide (recursion)

↓

Temp me sort (merge)

↓

Original me likh (copy back)

↓

Next level use updated array

ONE LINE TRUTH

👉 temp is temporary brain

👉 arr is permanent memory

Agar memory update nahi ki → pura system bhool jayega 🤖